Fiscal Year:	FY 2024	**Task Last Updated:**	FY 10/02/2023
PI Name:	Yodh, Arjun Ph.D.		
Project Title:	Crystal-to-Glass Transitions in Colloids (Flight Experiment)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-01
Start Date:	12/19/2018	**End Date:**	01/31/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
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Key Personnel Changes/Previous PI:	Ed. Note:No key personnel changes.		
COI Name (Institution):			
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Task Description:

This proposal builds on and continues the research described in Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 3) (Ed. note: see in Task Book "Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 3)"; grant NNX13AL27G ; Principal Investigator Arjun Yodh).
The roots of the work derive from an experimental program that was peer-reviewed and accepted for flight on the International Space Station (ISS); the instrumentation at the heart of the original flight proposal for ISS was part of the so-called Light Microscopy Module (LMM). More recently, a Science Concept Review (SCR) for Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 2) was held in June 2011, and the proposed research was strongly endorsed by the External Review Panel in their report on June 14 2011; the science goals and experimental plans for Low Volume Fraction Entropically Driven Colloidal Assembly (Phase 3), were also critically reviewed. The current research will be integrated into the hardware developed for the spaceflights that are named by variants of the term "ACE" (e.g., ACE-H1). In addition, related complex fluid measurements have been carried out in the Binary Colloid Alloy Test (BCAT) BCAT-series flight experiments. In response to these developments, the science aims have evolved to take advantage of important recent breakthroughs in the field. Herein we describe these complex fluid experiments. Note also, whenever possible, we will be joining forces with our colleagues from Europe to perform collaborative experiments within the European Space Agency (ESA) and NASA facilities.

The overall rationale for these investigations is that the elimination of particle sedimentation effects in microgravity creates a purely "thermodynamic" environment for colloidal suspensions wherein particle size, shape, and volume fraction, and interparticle interactions are the dominant determinants of the assembled structures. This feature enables us to make new materials that do not form on Earth and to elucidate the effects of gravity in soft matter science that are masked in ground-based experiments.

This version of Low Volume Fraction Entropically Driven Colloidal Assembly explores a range of experimental complex fluids with themes related to and evolved from previous proposal aims. The original studies were centered around entropically driven low volume-fraction self-assembling complex fluids, suspensions composed of an ensemble of large particles, and additional smaller constituents (either a small colloidal particle, a polymer, or a micelle). Phase 3 studies retained many of the original themes but also had new focus that reflected the evolution of the complex fluid field, as well as new developments in our laboratory. Broadly, the proposed research will elucidate a wide-range of phenomena in condensed matter physics that has implications for phase-diagrams of entropically driven colloidal media, crystallization, and melting mechanisms in three-dimensional (3D) bulk media and in finite volumes, jamming, and glass formation in 3D, and, opportunistically, crystallization and melting mechanisms and aging of colloids/soft-matter based on more anisotropic particles and filaments on a variety of length scales.

International collaborators are Prof. Luca Cipelletti, Université Montpellier II, Montpellier, France; Prof. Roberto Piazza, Department of Nuclear Engineering, Politecnico di Milano, Milan, Italy; Prof. Willem Kegel, Van 't Hoff Laboratory for Physical and Colloid Chemistry, Debye Research Institute, Utrecht University, Utrecht, The Netherlands.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

This research can provide fundamental understanding of natural phenomena affected by gravity for increased efficiency and effectiveness of space-based and industrial processes.

Task Progress:

Revised Objective: Before the ACE-T4 experiment on the International Space Station (ISS) was run, the failure of a master hard disk controlling the Light Microscopy Module (LMM) delayed starting this experiment by many months. Since the ACE-T4 experiment was designed to be "late-stow" to minimize the time between when the samples were created (concentrated) and when they were run in the LMM on ISS, the extended delay had a detrimental effect. The delay resulted in the gelation of the samples before the experiment began. The originally planned experiments were no longer possible due to gelation, but the ACE-T4 experiment based on new samples (gels) provided us with qualitatively different insights into the behavior of 3D colloidal gels over a range of temperatures and temperature gradients. Some effects were visible with LMM and provided an opportunity to study the microstructure, dynamics, and stresses in gels that can lead to microcracks, an important problem that, for example, could affect 3D printer inks.
Results (Flight Experiment): The ACE-T4 experiment (revised) used confocal microscopy to observe the gels over a range of particle packing-fractions (temperatures), using pure, monodisperse N-isopropyl acrylamide based polymer (pNIPAM) particles in one sample, and then mixtures of pNIPAM particles with different amounts of particle dopants in five other samples. All of the samples were observed to form gels. There was no crystallization in any sample in microgravity. The data we obtained permits the study of 3D gels with and without dopants as a function of temperature. The temperature changes vary the particle size; and thus, properties of the gels. We have been analyzing this data to see how the gel structure varies in the absence of sedimentation. To date, we can characterize these samples, but we do not as yet have a story to publish. It is possible, with further work on the ground and further analysis of the flight experiment data that we will develop a picture of the effects of microgravity on these sample types.

Results (Ground-based Research): In addition to the revised ACE-T4 experiment, we carried out and published important ground-based soft matter research. The published papers are described comprehensively in another section of this final report. [Ed. Note: See Bibliography.]

Here we very briefly summarize some of the important findings.

Colloid Research: We carried out important studies of supercooled liquids (glass precursors) and glasses. Notable work in this vein included a paper that experimentally correlated a new structural parameter called softness (which is related to the positions of neighbors surrounding a particle) with the time between particle rearrangements; if the time period is short, then the particle is "soft" and the local structure is amenable to change. Another paper experimentally tested predictions of the well-known inhomogeneous mode-coupling theory of glasses for the first time, especially power-law scaling of dynamic heterogeneity size and relaxation versus packing fraction. A third paper made experimental observations of supercooled colloidal liquids and glasses that suggested that short-time rearrangment events are very often precursors to heterogeneous relaxation at longer timescales. We also studied the ideas of excess entropy in thermal (equilibrium) glassy materials and in more granular disordered solids plastically deforming under shear. In this work, we experimentally investigated the relationship between the relaxation dynamics and microscopic structure of disordered colloidal solids during plastic deformation. In one paper, the experiments revealed that the relaxation rate associated with plastic flow at time t is correlated with the strain rate and sample microscopic structure measured at earlier and later times, respectively.

	Liquid Crystal (LC) Research: We carried out a variety of investigations of liquid crystals that discovered fundamentally new reconfiguration effects of the nematic director structure and suspended particles. In so-called lyotropic chromonic liquid crystals (LCLCs), we found that suspended particles induced local twist (twisted tails) of the director field in the vicinity of the particle. For spheres, this led to drastically different diffusion coefficients along different directions in the LCLC; and for rods, the twisted tails caused the rods to rotate to different orientation angles with respect to the far-field director. In a different vein, we studied LC drops in magnetic fields, demonstrating a novel point-to-ring first order topological defect transition. We also studied fluctuations near the core of radial hedgehog drops and new focal conic configurations of smectic LC Janus drops.
Bibliography Type:	Description: (Last Updated: 06/16/2025)
Articles in Peer-reviewed Journals	Wei WS, Jeong J, Collings PJ, Yodh AG. "Focal conic flowers, dislocation rings, and undulation textures in smectic liquid crystal Janus droplets." Soft Matter. 2022 May 17;18(23):4360-71. https://doi.org/10.1039/D1SM01623G , May-2022
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